Do Environmental Provisions in Trade Agreements Reshape Export Composition?

Extended summary

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1. The question

Nearly every trade agreement signed in the past two decades contains environmental language, and the number of such provisions has grown steadily. Whether any of it changes what countries actually trade is close to an open question. The evidence we have comes almost entirely from gravity models on aggregate bilateral flows, which can tell us whether trade between two countries grew, but not whether its composition shifted — whether a country began exporting more environmental goods and fewer pollution-intensive ones.

This paper asks that second question for the world's largest exporter. Between 2000 and 2015 China signed fourteen preferential trade agreements covering twenty-five destination economies, with environmental content ranging from a handful of generic cooperation clauses to a dedicated chapter. Did the agreements with more environmental content tilt Chinese exports toward green goods and away from dirty ones?

The answer is no on the green margin, with a bound attached, and inconclusive on the dirty margin. The more useful contribution is not the null itself but what produces it: the environmental content of these particular agreements consists almost entirely of language with no mechanism through which it could move a product mix. The null is what the existing literature's own heterogeneity results predict for a sample composed this way.

2. Setting and data

Trade data. The universe of Chinese customs transactions, 2000–2015: 45.8 million firm–HS6–destination–year observations covering 462,651 exporting firms. The granularity is what makes the design possible — the same firm can be observed shipping the same product to a destination covered by an agreement and to one that is not, in the same year.

Environmental content. Two independent codings, used in parallel throughout. The World Bank Deep Trade Agreements database gives a count of environmental provisions in force with each destination at each date, together with counts for the non-environmental policy areas. The TREND database codes environmental provisions at finer granularity, with sub-indices for green market access, enforcement, hard and soft obligations, and regulatory space. Using two codings guards against the idiosyncrasies of either.

Product classification. Green goods follow the OECD Combined List of Environmental Goods; dirty goods follow the standard pollution-intensity taxonomy mapping polluting industries to HS6 codes. Everything else is neutral and serves as the within-cell comparison group.

Samples. Hong Kong and Macao are excluded from the main estimates. Both have Closer Economic Partnership Arrangements with the mainland, and both are entrepôts: a large share of what the data record as Chinese exports to them is transshipped onward, so the recorded destination and the actual final market differ systematically. They account for 24.4 percent of treated observations and 50.1 percent of treated export value, and results including them are reported separately. After iterative removal of singleton observations the estimation sample is 21,519,511 observations with 225 destination clusters, 23 of them treated.

3. Empirical strategy

Why composition rather than levels

The obvious specification — regress log exports on environmental depth — cannot work here, for a structural reason rather than an empirical one. Any fixed-effect structure that takes bilateral confounding seriously requires a destination–year intercept, and that intercept absorbs everything that varies at the destination–year level, environmental depth included. With thirteen distinct agreement profiles and almost no within-destination change over time, "having deep environmental content" and "having a deep agreement" are empirically the same variable.

An interaction survives the destination–year intercept because it varies within the destination–year cell, across products. That is why the design is built on composition: it compares how green and dirty products respond relative to neutral ones, inside the same firm–destination–year.

The specification

$$\ln x_{fpdt} = \beta_1\, EP_{dt} \times g_p + \beta_2\, EP_{dt} \times b_p + \gamma_1\, TD_{dt} \times g_p + \gamma_2\, TD_{dt} \times b_p + \theta_{fpd} + \theta_{fdt} + \theta_{pt} + \varepsilon_{fpdt}$$

g_p and b_p are the green and dirty indicators, EP_{dt} environmental depth, TD_{dt} non-environmental agreement depth. The three sets of absorbed effects do different work. θ_{fpd} removes time-invariant features of each firm–product–market relationship. θ_{fdt} is what drives identification: it removes everything time-varying and bilateral — the agreement itself, demand shocks, exchange rates, the non-random selection of partners — leaving only the differential response of green and dirty products within the cell. θ_{pt} removes global product-level shocks such as the worldwide expansion of the solar market.

Because environmental depth and general agreement depth are highly correlated, every specification controls separately for non-environmental depth, and the paper reports every result under two independent measures of it: the World Bank count and the DESTA index. This turns out to matter, and Section 5 explains why.

Inference

Twenty-three treated destinations, sharing thirteen distinct agreement profiles, because the ten ASEAN partners enter through a single agreement. Under those conditions cluster-robust standard errors over-reject, sometimes badly. Every estimate is therefore reported with a wild cluster bootstrap p-value ($B = 9{,}999$), which is the reference inference, and the collapsed-panel estimates additionally with a permutation p-value.

The permutation test is worth describing because it does something the bootstrap does not. The environmental profile of each agreement — its depth and the years in which it takes effect — is detached from its destination and reassigned at random among the treated destinations; the model is re-estimated; the exercise is repeated a thousand times. The resulting p-value is the share of random reassignments producing a coefficient at least as large as the observed one. It answers the question: would this number have appeared if we had simply relabelled which of these countries signed environmental chapters?

4. Results

First, the fixed effects earn their keep

Estimating the same interactions under progressively richer fixed-effect structures produces a striking pattern. Under any structure that lacks an absorber with both a destination and a time dimension, the interactions are frequently positive and sometimes significant: the World Bank dirty interaction reaches $+0.0064$ ($p = 0.019$), the TREND green interaction $+0.0021$ ($p = 0.021$, column 3). Both margins moving in the same direction is not a composition effect — it is the growth of Chinese exports to partner countries around the entry into force of an agreement, loading onto whichever interaction happens to be correlated with agreement timing.

Adding firm–destination–year effects removes it. The World Bank dirty coefficient goes from $+0.0064$ to -0.0044 and the TREND green coefficient from $+0.0018$ to -0.0001, with the significance disappearing alongside the sign. We cannot identify which destination-specific shock was responsible, only that it had a destination–time structure, since absorbing that structure is what eliminates it.

The green margin: a bounded null

On the full panel the green interaction is -0.0023 (s.e. 0.0039) with the World Bank index and -0.0001 (0.0010) with TREND. Changing the depth control moves the third decimal. The joint F-test on all four composition interactions gives $p = 0.31$ and $p = 0.71$. Quantity and unit value are equally silent, which matters more than it appears: it rules out the possibility that a volume response and a price response are cancelling inside the value aggregate. Neither channel moves.

The null holds when the comparison group is replaced three separate times — comparing products only within the same HS4 heading, matching destinations on pre-treatment observables, and dropping untreated destinations altogether so that identification comes from differences in depth among partners. Across the twelve green cells of the three subsamples the coefficient stays between -0.0023 and $+0.0005$ and its bootstrap p-value never falls below 0.58; including the two reference rows, the range widens only to $[-0.0046, +0.0018]$ and the lowest p-value is 0.39.

The bound is what makes this more than a failure to reject. The 95 percent bootstrap interval runs from -0.0353 to $+0.0355$ per World Bank provision. Any effect larger than about three percent per provision is ruled out at 95 percent confidence; anything smaller, the design cannot see. The statement the paper defends is a bound, not an absence.

The dirty margin: negative everywhere, decisive nowhere

The dirty coefficient is negative in almost every specification and its magnitude is stable. Its statistical standing is not, and the reason is instructive.

Under the World Bank measure of general agreement depth the full-panel coefficient is -0.0044 with an asymptotic p of 0.052, which the bootstrap raises to 0.19. Under the DESTA measure the same coefficient is -0.0056 with asymptotic p of 0.039 and bootstrap p of 0.049. Same data, same fixed effects, same sample. What differs is collinearity: net of the fixed effects, environmental depth correlates 0.959 with the World Bank control and 0.891 with DESTA. The World Bank control absorbs nearly the same variation as the regressor of interest, leaving a wide standard error; DESTA overlaps less and the standard error narrows. Nothing about the estimated effect changes between the two — only how much of it the design can resolve.

The permutation test settles what the bootstrap cannot, because it reassigns whole agreement profiles rather than partialling out a control and therefore gives the same verdict under both depth measures. On the baseline it returns $p = 0.278$ for the dirty coefficient and $p = 0.597$ for green; under DESTA, 0.146 and 0.480. A coefficient that would arise a quarter of the time from randomly relabelling which of these destinations signed environmental chapters is not evidence about environmental content.

A leave-one-out exercise sharpens the diagnosis, and not in the usual way. The point estimate is remarkably stable, sitting between -0.0097 and -0.0133 across all 23 exclusions. What is fragile is precision: dropping Australia moves the coefficient by 13 percent but nearly triples its standard error, from 0.0030 to 0.0087; dropping Korea doubles it. These two destinations are not outliers pulling the estimate — they supply the variation that identifies it. And which of the two is pivotal depends on the depth control: under DESTA it is Korea rather than Australia.

Between firms or within them?

Aggregating the data to product–destination–year cells and re-estimating with the corresponding fixed effects yields a dirty coefficient of −0.0119 against −0.0044 in the firm-level panel, a factor of 2.7. The two specifications differ by exactly one thing — whether the firm dimension is in the absorbers — so the gap measures it: $(0.0119 - 0.0044)/0.0119 = 0.63$. Roughly three-fifths of the aggregate dirty association is a change in *which firms* export, not reallocation inside a given firm.

This reconciles the present result with recent work on Chinese customs data reporting that environmental provisions reduce dirty exports and raise green ones. That literature identifies from variation across firms and destinations; the specification here removes it. The two are not in conflict once the estimands are stated.

Table 1: The result in one table (log export value, World Bank depth control)

	WB EP depth		TREND EP count	
Sample / design	× green	× dirty	× green	× dirty
Full panel (reference)	−0.0023	−0.0044	−0.0001	−0.0009
bootstrap p	[0.686]	[0.185]	[0.931]	[0.177]
Collapsed panel	−0.0046	−0.0119	+0.0018	+0.0004
bootstrap p	[0.649]	[0.072]	[0.390]	[0.857]
permutation p	[0.597]	[0.278]	[0.160]	[0.817]
C-prod-HS4	−0.0009	+0.0066	+0.0005	+0.0024
bootstrap p	[0.860]	[0.435]	[0.730]	[0.734]
CEM-matched destinations	−0.0023	−0.0041	−0.0006	−0.0011
bootstrap p	[0.610]	[0.251]	[0.609]	[0.204]
Deep vs. shallow (partners only)	−0.0022	−0.0034	−0.0004	−0.0006
bootstrap p	[0.585]	[0.499]	[0.751]	[0.739]

Notes: Each cell is a separate estimation. Hong Kong and Macao excluded throughout; all specifications control for non-environmental World Bank depth interacted with both product types. Wild cluster bootstrap, $B = 9{,}999$, clustered by destination. Permutation p-values reported for the collapsed panel, where the treated-only design is feasible. No stars: with this many designs, star-counting invites exactly the selective reading the table is meant to prevent. The equivalent table under the DESTA depth control is in the paper; the only cells that move are the dirty ones.

Does the content of the clauses matter?

The natural objection to any null on an aggregate index is dilution. The indices add up every environmental provision regardless of what it does, and most of what they add up is cooperation language with no channel to a product mix.

The dilution is severe. Of 150 World Bank provision slots checked across the treated destinations, 8 — 5.3 percent — fall in sub-indices with a trade mechanism. In TREND it is 4 of 437, or 0.9 percent. A binding environmental obligation appears in exactly one of the fourteen agreements, the 2015 agreement with Korea. The two World Bank sub-indices with a trade mechanism are non-zero in only three destination–years and always in the same proportion, which makes them perfectly collinear and impossible to separate.

Replacing the aggregate index with one sub-index at a time reproduces the aggregate pattern: nothing on the green margin anywhere, and on the dirty margin a marginally negative coefficient on enforcement provisions that rests on asymptotic inference alone.

The conclusion is negative and it is the honest one. This sample cannot distinguish between "there is no effect" and "there are no provisions capable of producing one". What it can establish is that the two are observationally equivalent here.

Dynamics and the extensive margin

Pre-treatment coefficients in the event study are flat on both margins. A mild negative drift for green products late in the post-period is identified only by the cohorts that entered by 2010, dominated by the two early ones (Bangkok 2002 and ASEAN 2005), and disappears under the Sun–Abraham interaction-weighted estimator, whose aggregated effects are -0.042 ($p = 0.27$) on green and $+0.073$ ($p = 0.28$) on dirty. Allowing destination-specific trends over the full sample period turns the TREND green interaction significantly negative — the only green coefficient in the robustness battery to survive the bootstrap — but the green gap was already rising before entry into force, which makes a pre-existing trend the more economical account.

Poisson estimation on a zero-filled product–destination–year grid finds no green trade creation at the extensive margin: $+0.0015$ ($p = 0.74$) with the World Bank index and $+0.0001$ ($p = 0.95$) with TREND.

5. Interpretation

The result is not that environmental provisions cannot affect trade. It is that these provisions, in these agreements, did not.

The literature that finds effects finds them in the same place. Brandi et al. (2020) report that only provisions with a direct trade mechanism move anything: liberal provisions raise green export shares, restrictive ones reduce dirty exports, and general cooperation language does nothing.

Abman et al. (2024) find that forest-preservation clauses offset the deforestation that typically follows an agreement — again, a specific clause with an identifiable mechanism. China's agreements of this period are, on the same coding, overwhelmingly cooperation-only.

So the null is the predicted outcome of the existing literature's own heterogeneity findings, applied to a sample where the treatment lacks the specificity that drives effects elsewhere. The policy reading is direct: a chapter in an agreement is not the instrument. What the chapter contains is.

There is also a methodological contribution, arrived at reluctantly. With treatment concentrated across roughly a dozen agreements, an asymptotically overwhelming composition effect ($p < 0.001$) can dissolve entirely under bootstrap and permutation inference. Applying those tests as the primary mode of inference rather than as a robustness appendix should be standard for this class of design.

6. What the paper cannot say

Four limitations, stated plainly.

Levels. The design speaks only to composition. Whether agreements with more environmental content changed aggregate trade with those partners is not identified here, and cannot be: a destination–year intercept absorbs the question along with the variable.

The dirty margin is genuinely unresolved. The coefficient is negative almost everywhere and its statistical standing depends on a modelling choice. We read it as indistinguishable from a placebo on the strength of the permutation test, but a dose–response interpretation — the effect is real and concentrated in the two destinations with the deepest environmental coverage — is not ruled out by the evidence assembled here.

Provision content. With thirteen distinct agreement profiles and perfect collinearity among the trade-mechanism sub-indices, the claim that content matters is an inference across studies rather than something this design demonstrates.

Tariffs and firm entry. The preferential tariff rate was not obtainable from the customs data; the MFN rate serves as a control in the robustness battery only, and the headline estimates do not use it. And the extensive margin is probed at the product–destination level: a new firm beginning to export a green product that other Chinese firms already ship to that destination is not detected.